

Downloading the Required Packages

Human Development Index (HDI) Prediction Using Machine Learning

Introduction

Before developing the HDI Prediction System, the required Python libraries and development tools must be installed. These packages support data analysis, machine learning, visualization, and web application development.

Required Software

- Python 3.10 or later
- Visual Studio Code (VS Code) or PyCharm
- Git
- GitHub

Required Python Packages

Package	Purpose
NumPy	Numerical computations
Pandas	Data analysis
Matplotlib	Visualization
Seaborn	Statistical graphs
Scikit-learn	Machine learning
Flask	Web application
Joblib	Save/load trained models

Installation Commands

```
python -m pip install --upgrade pip
```

```
pip install numpy pandas matplotlib seaborn scikit-learn flask joblib
```

Verification

Run **pip list** to verify that all required packages have been installed successfully.

Expected Outcome

After installation, the environment is ready for data preprocessing, machine learning model training, visualization, and deployment of the HDI Prediction System.